1. FIND THE REMAINDER OF THE INDICATED DIVISION BY THE REMAINDER THEOREM.

$(x^4 - 5x^3 + 8x^2 + 15x - 2) \div (x - 3)$

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2. USING THE FACTOR THEOREM IF $t+5$ IS A FACTOR OF

$f(t) = 2t^4 - 10t^3 - t^2 - 3t + 10$

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3. USE SYNTHETIC DIVISION TO PERFORM $(x^6 + 63x^3 + 5x^2 - 9x - 8) \div (x+4)$

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4. FIND ALL ROOTS OF $2x^4 - 2x^3 - 10x^2 - 2x - 12 = 0$

GIVEN THE TWO ROOTS $r_1 = 3,\ r_2 = -2$

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5. FIND THE ROOTS OF $S^5 + 3S^4 - S^3 - 11S^2 - 12S - 4 = 0$

IF -1 IS A TRIPLE ROOT

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6. SOLVE $6y^3 + 19y^2 + 2y = 3$

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7. IF A CALCULATOR GRAPH SHOWS A REAL ROOT, HOW MANY NON REAL COMPLEX ROOTS ARE POSSIBLE FOR A SIXTH-DEGREE POLYNOMIAL EQUATION $f(x) = 0$

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8. FIND RATIONAL VALUE OF $\underline{a}$ SUCH THAT $(x - a)$ WILL DIVIDE INTO $x^3 - 13x + 3$ WITH A REMAINDER OF -9

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(9) FIND THE VALUE OF K TO HAVE $x = 3$ IS A ROOT FOR THE FUNCTION $f(x) = Kx^4 - 15x^2 - 5x - 12$

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(10) IF $f(x) = 3x^4 - 18x^3 - 2x^2 + 13x - 6$ AND $f(x) = g(x)(x-6)$, FIND $g(x)$

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(11) A CUBICAL TABLET FOR PURIFYING WATER IS WRAPPED IN A SHEET OF FOIL 0.5 mm THICK. THE TOTAL VOLUME OF TABLET AND FOIL IS 33.1% GREATER THAN THE VOLUME OF THE TABLET ALONE. FIND THE LENGTH OF THE EDGE OF THE TABLET.

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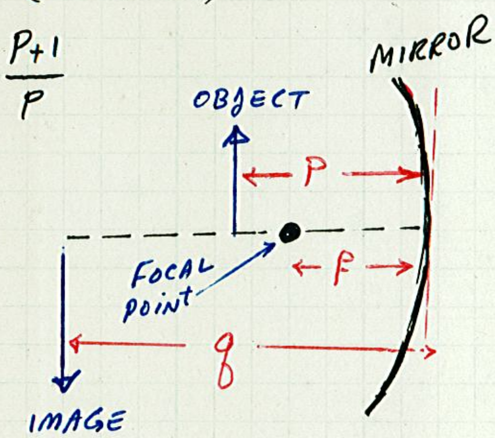
(12) FOR THE MIRROR SHOWN, RECIPROCAL OF THE FOCAL DISTANCE f EQUALS THE SUM OF RECIPROCALS OF THE OBJECT DISTANCE p AND THE IMAGE DISTANCE q (IN INCHES).

FIND p, IF $q = p + 4$ AND $f = \dfrac{p+1}{p}$



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(13) THE RADIUS OF ONE BALL BEARING IS 1.0 mm GREATER THAN THE RADIUS OF A SECOND BALL BEARING. IF THE SUM OF THEIR VOLUMES IS 100 mm^3, FIND THE RADIUS OF EACH.
